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FH Aachen

Faculty of Aerospace Engineering

Space Flight Mechanics

Bachelor Thesis

Usage of Aerobraking in the Saturnian System

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1 Introduction

Please add your introductory text here.

2 Description of the program code

2.1 User Input

The user input is the first region of the program and is used to modify the simulation to be as close to the planned mission as possible. It is divided into the three categories Rocket parameters, Orbital parameters, and Output parameters. This part begins with "region User input" and ends with "endregion".

Constant	Name in program	Value	Unit
Total mass	`m_0`	3725.0	kg
Propellant mass	`m_p`	2725.0	kg
Thrust	`thrust`	0	N
Hyperbolic excess velocity	`v_inf`	1 .. 3	km/s
Specific Impulse	`Isp`	3100	s
Drag Coefficient	`c_d`	2.2	–
Cross-Sectional Area	`A`	29.3	m^2
Heat Load Coefficient	`c_q`	0.5	–

Table 2.1: Rocketparameters (own figure)

Constant	Name in program	Value	Unit
Simulation duration	`tint`	540	days
Lowest periapsis	`r_p_lowest`	258	km
Highest periapsis	`r_p_highest`	260	km
Step size	`r_p_step_size`	0.1	km
Periapsis of desired Orbit	`r_p_orbit`	1500	km
Apoapsis of desired Orbit	`r_a`	2000	km

Table 2.2: Orbital parameters (own figure)

Constant	Name in program	Options
Simulation duration	`Plot_Trajectory`	True/False
Lowest periapsis	`Plot_Atmosphere`	True/False
Highest periapsis	`Plot_Values`	True/False
Step size	`Optimize_Thrust_Duration`	True/False
Periapsis of desired Orbit	`Plot_Heating_Rate`	True/False

Table 2.3: Outputparameters (own figure)

2.2 Preparing the program

This section of the code begins with "region Prepare the program" and ends with "endregion". It consists of general planetary and orbital constants, accuracy parameters for the simulation, Unit conversion as well as initializing variables to track the spacecrafts state during the simulation.

Constant	Name in program	Value	Unit
Gravitational Acceleration	`g0_earth`	9.80665	m/s^2
Gravitational Parameter of Saturn	`mu_saturn`	37931207.8	km^3/s^2
Radius of Saturn	`R_saturn`	60268.0	km
Distance between Saturn and Titan	`dist_saturn_titan`	122187.0	km
Saturn's Sphere of Influence Radius	`R_SOI_Saturn`	54500000	km
Gravitational Parameter of Titan	`mu_titan`	8978.14	km^3/s^2
Radius of Titan	`R_titan`	2574.73	km
Titan's Sphere of Influence Radius	`R_SOI_titan`	1200000	km
Titan's semi-major axis	`a_titan`	1221870	km
Eccentricity of Titan's Orbit	`e_titan`	0.0288	–
Orbital Period of Titan	`T_titan`	15.945	days
Gravitational Parameter of Enceladus	`mu_titan`	7202	km^3/s^2
Radius of Enceladus	`R_titan`	252.1	km
Enceladus' semi-major axis	`a_titan`	237948	km
Eccentricity of Enceladus' Orbit	`e_titan`	0.0047	–
Orbital Period of Enceladus	`T_titan`	1.370218	days

Table 2.4: Planetary and Orbital constants (own figure)

2.3 Atmospheric Model

The atmospheric model is a standard exponential model and extends up to 1375km altitude. The base value at the surface level is 5.38 $\mathrm{kg/m^3}$. At its farthest point, this value decreases to 1e-11 $\mathrm{kg/m^3}$. The atmospheric model is based on the data from the Huygens Atmospheric Structure Instrument. A more in-depth comparison is found in 3.2.1 (VERLINKEN)

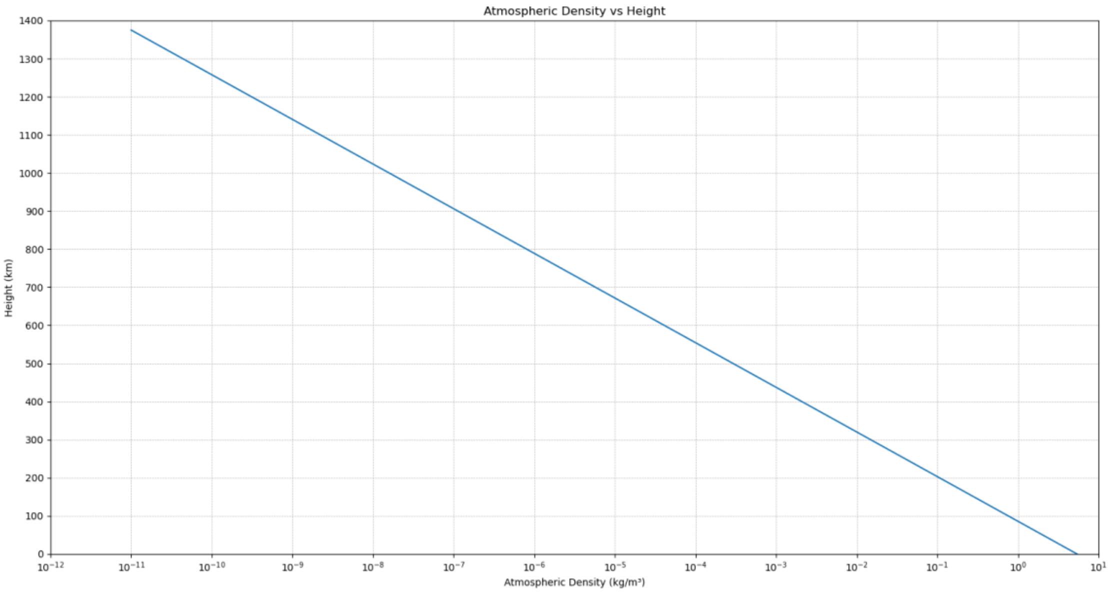

Figure 2.1: Atmospheric Density Model

2.4 Mission phases

2.5 Equations of Motion

The equations of motion section of the code contains four functions. The three helper functions "get_titan_position", "get_enceladus_position", "get_initial_spacecraft_state and the main function that calculates the following parameters of the spacecraft for each point in the simulation.

Parameter	Name in program	Unit
Velocity	`v`	km/s
Flight path angle	`alpha`	rad
Specific orbital energy	`E`	MJ/kg
Specific angular momentum	`h`	km/s^2
semi-major axis	`a`	km
eccentricity	`e`	–
apoapsis	`apoapsis`	km
periapsis	`periapsis`	km
descending	`is_descending`	True/False
throttle	`throttle`	0/1
atmospheric density	`atmo_density`	kg/km^3
Airdrag	`D`	$kg * km/s^2$

Table 2.5: Outputparameters (own figure)

The following five differential equations are solved in the main function and determine the spacecraft status parameters listed above in Table 2.5 at each simulation time step.

$$\frac{d\mathrm{r}}{d\mathrm{t}} = \rho_{\mathrm{r}} \tag{2.1}$$

$$\frac{d\phi}{d\mathrm{t}} = \rho_{\phi} \tag{2.2}$$

$$\frac{d\rho_{\mathrm{r}}}{d\mathrm{t}} = \mathrm{r}\,\rho_{\phi}^{2} - \frac{\mu}{\mathrm{r}^{2}} - \frac{\mathrm{D}}{\mathrm{m}}\cos(\alpha) + \frac{\mathrm{throttle}\cdot\mathrm{thrust}}{\mathrm{m}}\cos(\beta) \tag{2.3}$$

$$\frac{d\rho_{\phi}}{d\mathrm{t}} = -2\,\rho_{\mathrm{r}}\,\rho_{\phi} - \frac{\mathrm{D}}{\mathrm{m}}\sin(\alpha) + \frac{\mathrm{throttle}\cdot\mathrm{thrust}}{\mathrm{m}}\sin(\beta) \tag{2.4}$$

$$\frac{d\mathrm{m}}{d\mathrm{t}} = -\frac{\mathrm{throttle}\cdot\mathrm{thrust}}{\mathrm{c_{eff}}} \tag{2.5}$$

2.6 Launch Delay Calculation

The Launch Delay Calculation simulates the trajectory of the spacecraft until it crosses Titans Orbit the first time and checks for the angular difference between the spacecraft and Titan at that time. It aims to find a launch delay that results in the spacecraft passing in front of Titan at the distance given in the User Input region of the program. It utilizes a database as starting values made from delay values that have proven to result in fast convergence of the iteration process. The iteration process follows a distinct logic of checking for the angular difference first and adjusting the delay according to that. If the angular difference is positive, since we plan to pass in front of Titan and 2 ° the adjustment will be based on the error to the target crossing distance. Every adjustment also includes the hyperbolic excess velocity as a factor. When a suitable delay time is found, the program moves to the next section to adjust the Aerobrake Altitude.

2.7 Adjust Aerobrake Altitude

This function utilizes the launch delay that was found by the Launch Delay Calculation. It then runs a longer simulation with that delay and checks for the second crossing of Titan Orbit and checks for the angular difference between the spacecraft and Titan.

3 Results

Analyzing possible aerobraking altitudes to stay within the Saturnian system results in finding multiple possible heights for every hyperbolic excess velocity up to 3 $\mathrm{km/s}$. Higher aerobraking altitudes reduce the speed of the spacecraft less and therefore result in a longer time until the next aerobraking maneuver and a longer mission duration overall. However, there is an advantage in another area, because the heat load on the spacecraft at heights above 600 km for the first crossing is significantly lower. In fact, low enough so that no extra heat shielding is necessary and radiative heat dissipation using coatings is sufficient.

3.1 Simulation results

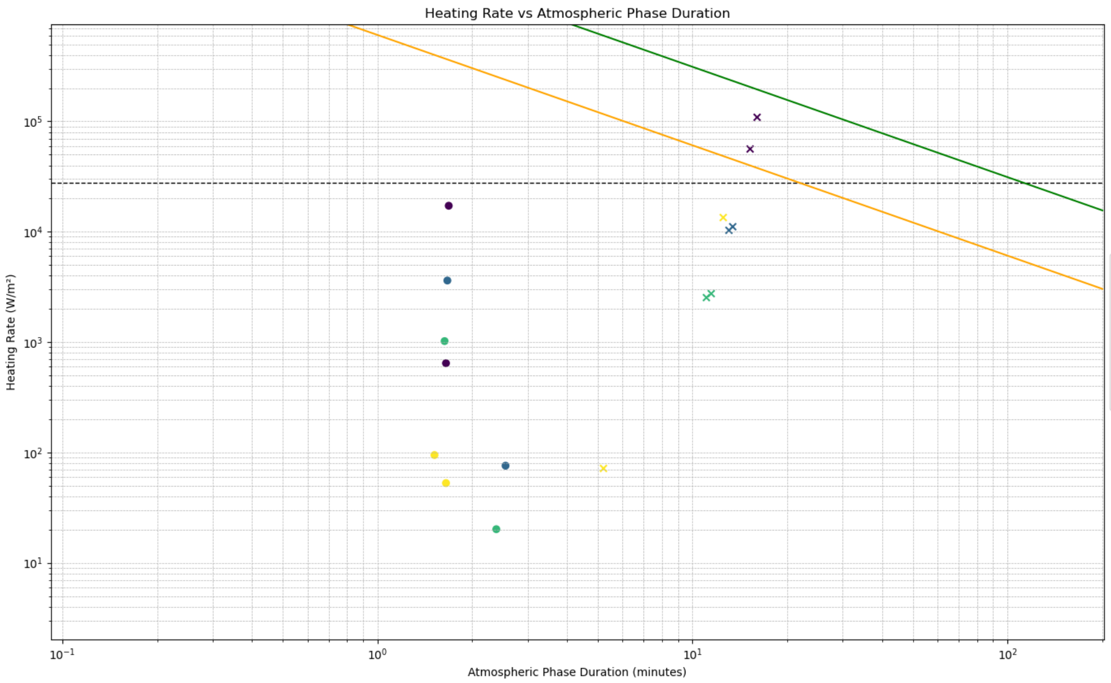

Figure 3.1: Heating Rate

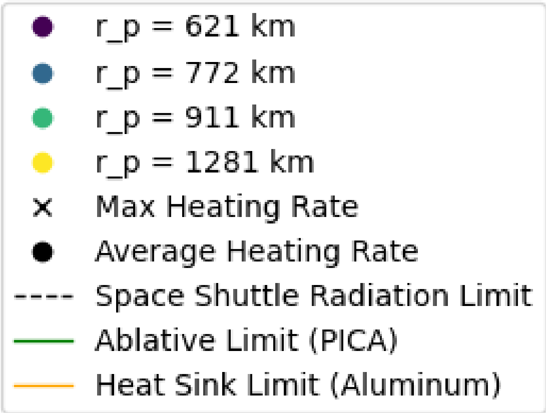

Figure 3.2: Heating Rate Legend

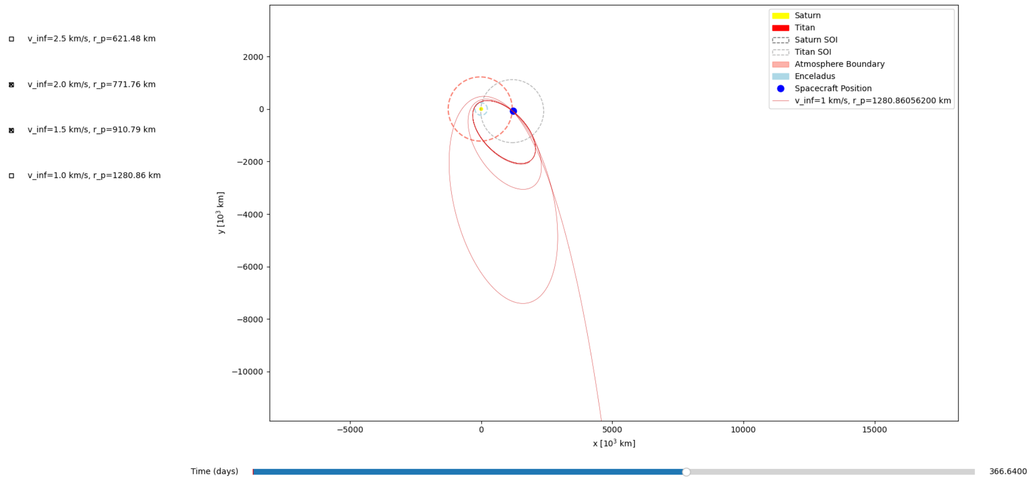

Figure 3.3: Trajectory Example

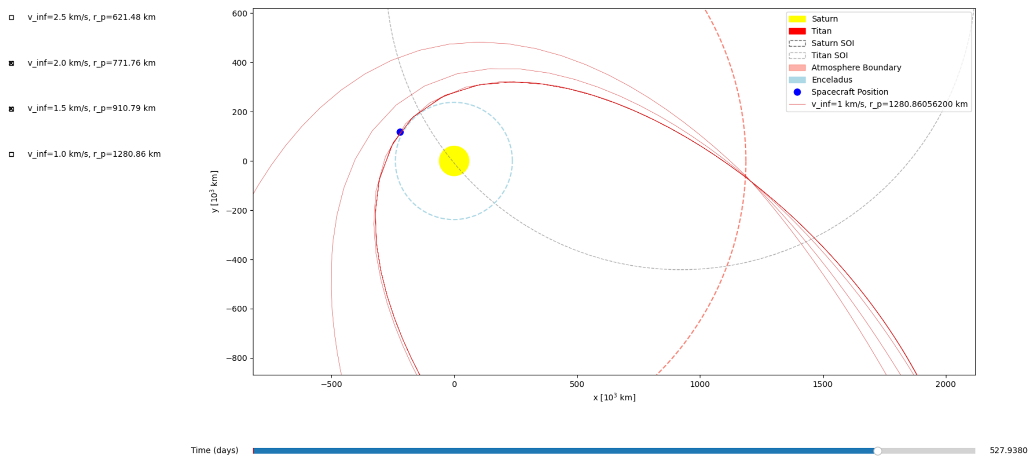

Figure 3.4: Trajectory Example detailed

3.2 Validation of results

3.2.1 Validation of the atmospheric model

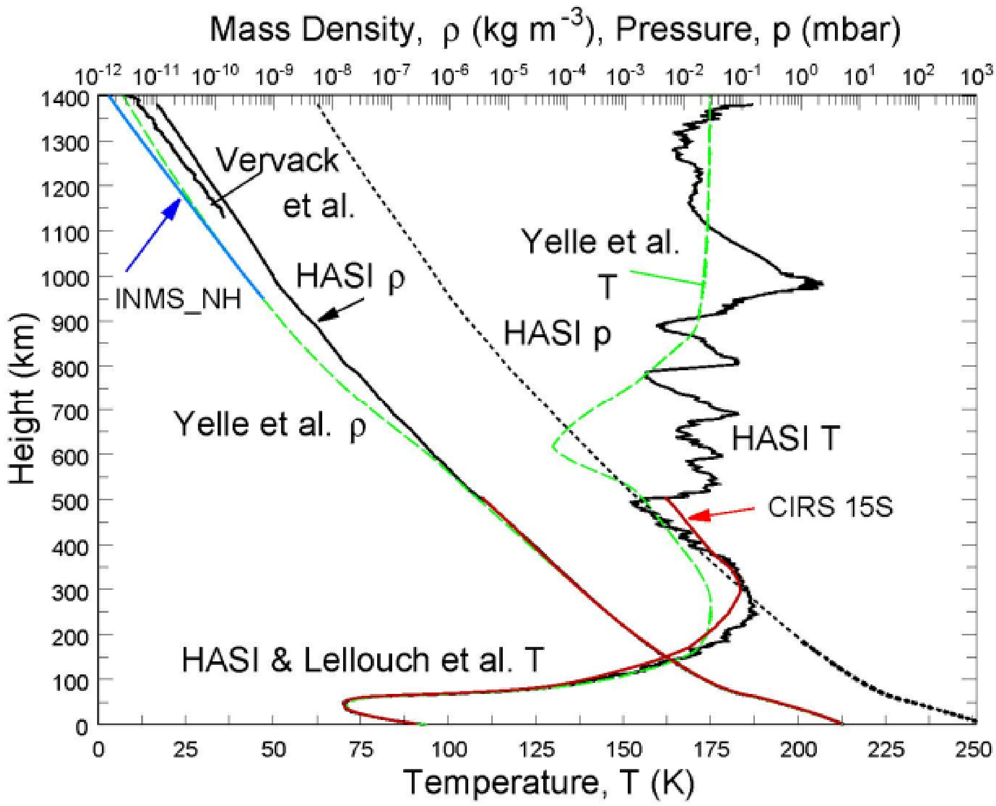

Figure 3.5: Atmospheric Density Data from HASI

3.2.2 Validation of aerobraking results

4 Conclusion

A Appendix Title

A.1 A First Section in the Appendix